Model Dependence of PDG Comparison Values:

How Empirical Values Arise, Where Theory and Irrational Factors Enter, and What This Means for FFGFT Residuals

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Contents

Status markers

[K] *Confirmed* — derived from ξ, numerically verified.

[B] *Proved* — algebraically established.

[S] *Speculative* — open.

[Q] *Source* — corpus-external primary source, verified here.

Abstract

FFGFT predictions are compared against PDG and CODATA values. This document examines how those comparison values actually arise: which raw observable underlies each, which theory (QED, Standard Model) enters the extraction, and where irrational numbers already appear in the extraction formulae themselves.

Main findings: (1) m_e and m_μ/m_e are the least model-dependent anchors, but m_μ/m_e is QED-dependent and derives from a 27-year-old muonium HFS measurement whose theory uncertainty CODATA underestimates by a factor of 5–10 (Eides 2026). (2) α carries an unresolved 5σ tension between the two best recoil measurements. (3) $v = 246$ GeV is never measured; it is the SM definition $v \equiv (\sqrt{2}G_F)^{-1/2}$. (4) G_F carries a theory correction $\Delta q = -0.44\,\%$ with a 4.7σ hadronic tension. (5) m_τ has two measurement methods differing by 0.18 MeV; FFGFT lies 0.4σ from the PDG average. (6) The Galois identity $(m_\mu/m_e)^2 = 43200$ and the PDG masses agree very well: the ratio deviates by $-38.99 \cdot \xi \approx 0.52\,\%$ linearly (resp.

$1.03\,\%$ quadratically). No known error source accounts for this residual — it is an empirical discrepancy not explained by the stated comparison uncertainty. The cause — non-uniform generation-dependent corrections ε_i from QED extraction and SI projection — is derived quantitatively in Doc. 352, § 9–10 (explanatory fraction 101 % **[K]**); the falling structure $\varepsilon_e > \varepsilon_\mu > \varepsilon_\tau$ itself remains open **[S]** (R113 bridge). (7) Errors accumulate along the derivation chain; every prediction requires its full provenance path.

All 21 primary sources were verified on 5 September 2026.

1 The question

The corpus compares FFGFT predictions with PDG values and finds residuals at the sub-percent to percent level. Are PDG values theory-neutral, or do they themselves depend on the Standard Model — such that part of the residual arises on the extraction side rather than the FFGFT side?

The FFGFT foundational statement: only ratios free of irrational numbers are exact; SI absolute values are structurally approximate (Docs. 241, 085). If PDG values themselves contain irrational factors in their extraction, the same limitation applies to the comparison side.

2 How empirical values arise

No PDG or CODATA value is a raw measurement. Each passes through the same chain:

raw observable → theory formula → extracted parameter → CODATA adjustment → PDG value

The CODATA adjustment

CODATA publishes recommended values every four years, produced by a joint least-squares adjustment over roughly 300 input data linked by theory relations (QED, Rydberg formula, kinematics). The values are therefore *correlated*: a change in one input shifts all dependent values. Where theory uncertainties enter, they are partly treated as adjustment parameters (δ_{th} in the observational equations) rather than as errors — this is the Eides finding for muonium HFS **[Q]**.

Electron mass m_e

Raw observable: cyclotron frequency of a hydrogen-like ion (^{12}C^{5+}, ^{28}Si^{13+}, ^{20}Ne^{9+}) in a Penning trap, and the Larmor precession frequency of the bound electron. The ratio is a pure number.

Theory formula: bound g-factor from QED bound-state theory. From the frequency ratio and g_{bound} follows m_e/m_{ion}; with the ion mass in u follows m_e in u; conversion to MeV via u → kg → MeV.

Theory dependence: QED for g_{bound}, small and well controlled. Current: Heiße et al. (2023) at 0.1 ppb **[Q]**.

Muon-electron mass ratio m_μ/m_e

Raw observable: ground-state hyperfine splitting of muonium (a bound μ^+e^- state) — a microwave resonance at 4.463 GHz — and Zeeman splitting in a magnetic field.

How m_μ/m_e is measured. The muon is not placed on a scale. From the measured HFS frequency $\Delta\nu$ and the QED formula for the Fermi frequency

$$\nu_F = \tfrac{16}{3}\,\alpha^2\,c\,R_\infty\,\frac{m_e}{m_\mu}\Big(\frac{m_r}{m_e}\Big)^3 \tag{1}$$

(corrections: QED to $O(\alpha^3)$, recoil in m_e/m_μ, weak Z exchange, hadronic VP) m_μ/m_e follows. Two consequences:

- m_μ/m_e *is not theory-free*: the extracted value contains assumptions about α, higher-order QED loop corrections, and hadronic vacuum polarisation. Since $\nu_F \propto \alpha^2$, α enters implicitly — anchor and α are not independent.
- *CODATA overstates the real precision*: CODATA 2022 gives $\sigma_{\text{rel}}(m_\mu/m_e) = 2.2\times10^{-8}$. The actual uncertainty of the HFS extraction (experiment + QED theory from Eides 2026 + α tension) is $\approx 1.2\times10^{-7}$ — a factor of 5–6 larger. **[Q]**

Theory dependence: high. Sole precision measurement: LAMPF 1999 (Liu et al.). Eides (2026) shows that CODATA lists the measured value 4 463 302 776(51) Hz as the theoretical value; the actual theory uncertainty is 271–515 Hz **[Q]**.

Fine-structure constant α

Two routes that disagree at 5σ **[Q]**:

- **Atom recoil:** $\alpha^2 = 2R_\infty\,(h/m_{\text{atom}})\,(m_{\text{atom}}/m_e)/c$. Rb (Morel et al. 2020): $\alpha^{-1} = 137.035\,999\,206(11)$. Cs (Parker et al. 2018): $137.035\,999\,046(27)$. Difference 1.6×10^{-7}, $> 5\sigma$.
- **Electron $g-2$:** a_e at 10^{-13} (Fan et al. 2023) plus 5-loop QED (12 672 diagrams, Aoyama et al.). Gives $\alpha^{-1} = 137.035\,999\,166(15)$ — between Rb and Cs.

The CODATA value 137.035 999 177(21) is an average covering none of the measurements. FFGFT: $\alpha^{-1} = 3700/27 = 137.037\,037\ldots$ (Doc. 338, R101), deviation 7.6 ppm — 100× larger than the Rb/Cs tension.

Tau mass m_τ — two methods

Method	Experiment	Value (MeV)	Theory assumption
Threshold scan	BESIII 2014	$1776.91 \pm 0.12\,^{+0.10}_{-0.13}$	QED cross-section
Pseudomass	Belle II 2023	$1777.09 \pm 0.08 \pm 0.11$	$m_{\nu_\tau} = 0$
PDG average 2024	—	$\mathbf{1776.93 \pm 0.09}$	—

Threshold scan: $\sigma(e^+e^- \to \tau^+\tau^-)$ versus beam energy near $2m_\tau \approx 3.55$ GeV; m_τ as fit parameter of the QED threshold curve. **Pseudomass:** at 10.6 GeV, the kinematic edge of the invariant mass in $\tau \to 3\pi\nu$; assumes $m_{\nu_\tau} = 0$.

For FFGFT: $m_\tau = 1776.97$ MeV (pred) lies 0.04 MeV $= 0.4\sigma$ from the PDG average. The methods differ by 0.18 MeV. The falsification test is undecided; Belle II with > 1 ab^{-1} should reach $\sigma < 0.05$ MeV.

Fermi constant G_F and vacuum expectation value v

Raw observable for G_F: muon decay time distribution (MuLan, 10^{12} decays), $\tau_\mu = 2.196\,980\,3(22)$ µs **[Q]**.

Theory formula (Eberhart et al. 2026 **[Q]**):

$$\tau_\mu^{-1} = \frac{G_F^2\,m_\mu^5}{192\pi^3}\,(1+\Delta q), \qquad \Delta q = (-4\,384\,678 \pm 34)\times10^{-9} \approx -0.44\,\%. \tag{2}$$

Δq contains QED to $O(\alpha^3)$, hadronic vacuum polarisation, electron-mass effects. Purely electroweak corrections are absorbed into G_F by definition (Sirlin 1980). The hadronic component carries a 4.7σ tension (CMD-3 vs. older data).

Raw observable for v: none. $v \equiv (\sqrt{2}\,G_F)^{-1/2} = 246.22$ GeV follows from $M_W = gv/2$ and $G_F/\sqrt{2} = g^2/(8M_W^2)$. v **exists only within the Standard Model.** Its numerical value is the muon lifetime in SM clothing.

For FFGFT: Doc. 006/R104 gives $v = 248.3$ GeV (without K_{frak}) or 245.6 GeV (with $K_{\text{frak}} = 74/75$). Comparison with 246.22 GeV is a comparison of *FFGFT structure against SM definition*, not against an observable. Only τ_μ and m_μ are measured.

Gravitational constant G

Torsion balance (Li et al. 2018, 11.6 ppm) or atom interferometry (Rosi et al. 2014, 150 ppm). Theory: Newton. The only model-free value, but the least precise: CODATA $6.67430(15)\times10^{-11}$, 22 ppm **[Q]**.

Neutrino mass-squared differences

Event rates versus L/E, three-flavour fit (NuFIT 6.0 **[Q]**). Δm^2_{31} depends on whether SK atmospheric data are included: ratio 33.7 (with) vs. 33.9 (without). FFGFT $360/11 = 32.73$; measurement uncertainty 1–3 % $\approx$ FFGFT residual. Not discriminating.

Overview

Quantity	Raw observable	Theory in extraction	Convention
m_e	frequency ratio	QED (g_{bound})	—
m_μ/m_e	HFS frequency	QED, weak, hadr.	—
α (Rb/Cs)	interferometer phase	R_∞ (QED)	—
α ($g-2$)	frequency ratio	5-loop QED	—
m_τ	rates vs. E / edge	QED threshold / $m_{\nu_\tau} = 0$	beam calibration
G_F	decay times	Fermi + QED + hadr.	Sirlin absorption
v	—	—	$v \equiv (\sqrt{2}G_F)^{-1/2}$
G	angle / acceleration	Newton	—
Δm^2	rates vs. L/E	3-flavour	with/without SK-atm

No value is theory-free. For FFGFT this means: comparison with PDG values is comparison with “QED + convention”, not with raw observations.

3 Irrational factors in the extraction formulae

The FFGFT foundational statement — ratios of rationals are exact, SI absolute values carry the approximation of the projection $S^1_m \to \mathbb{R}_t$ (Docs. 306, 333) — applies to the SM extraction formulae themselves:

Quantity	Formula	Irrational factor
G_F	$\tau_\mu^{-1} = G_F^2 m_\mu^5/(192\pi^3)$	π^3
v	$v = (\sqrt{2}\,G_F)^{-1/2}$	$\sqrt{2}$
α (recoil)	$\alpha^2 = 2R_\infty h/(mc)$	$R_\infty \propto \alpha^2$: implicitly circular
HFS	$\nu_F \propto \alpha^2$	α^2
Δq	expansion in α/π	π in every term

"G_F measured" means: τ_μ measured, divided by $192\pi^3$, square root taken. "v measured" additionally means: divided by $\sqrt{2}$, square root again. The irrational factors come from the SM formulation (phase-space integral, Higgs normalisation), not from measurement. The only convention-free comparison would be at the level of $\tau_\mu = 2.196\,980\,3$ µs — which FFGFT would have to predict directly **[S]**.

4 The residual in $(m_\mu/m_e)^2$: exact computation

FFGFT bare (Doc. 338): $(m_\mu/m_e)^2 = 43200$ exact, integer, $= |\mathrm{GF}(9)^*|^2 \cdot 5^2 \cdot |\mathrm{GF}(27)|$. With CODATA 2022 in exact rational arithmetic:

$$(m_\mu/m_e)^2_{\mathrm{PDG}} = 42753.12 \tag{3}$$

$$\text{Residual} = 446.88 \quad (1.045\,\%). \tag{4}$$

Possible error sources, computed:

- CODATA uncertainty $(m_\mu/m_e)^2$: ± 0.002 — residual remains $240\,000\times$ larger.
- Eides correction of HFS theory uncertainty (515 instead of 51 Hz): ± 0.010 — residual remains $45\,000\times$ larger.
- α choice in HFS anchor (FFGFT $3700/27$ instead of CODATA): shifts $(m_\mu/m_e)^2$ by -1.3 — residual remains $340\times$ larger.
- To reach 43200, m_μ would need to be 551 keV higher (measurement error 2.3 eV), or m_e 2.65 keV lower (measurement error 0.15 meV), or the HFS 23 MHz higher (measurement error 51 Hz).

Conclusion [B]: No measurement error explains the residual. The discrepancy is an empirical one not accounted for by the stated uncertainty of the comparison value. The cause — non-uniform ε_i from QED extraction (R113) and SI projection — is derived in Doc. 352, § 9–10 **[K]**; the falling structure $\varepsilon_e > \varepsilon_\mu > \varepsilon_\tau$ remains open **[S]** (R113 bridge). The difference axis $\varepsilon_\mu - \varepsilon_e \approx -39 \cdot \xi$ measures the ratio residual, the product residual $\approx -1.2 \cdot \xi$ the sum axis. Both are empirically computed **[K]**.

Two orthogonal axes: difference and sum of logarithms

The PDG masses m_e and m_μ produce two independent logarithmic projections when compared with the Galois identities:

$$\underbrace{\ln\frac{m_\mu/m_e}{\sqrt{43200}}}_{\text{difference axis}} = \varepsilon_\mu - \varepsilon_e \approx -38.99 \cdot \xi \qquad \underbrace{\ln\frac{m_e \cdot m_\mu}{54\,\mathrm{MeV}^2}}_{\text{sum axis}} = \varepsilon_e + \varepsilon_\mu \approx -1.21 \cdot \xi$$

Difference axis: linear and square are one single finding. The linear ratio m_μ/m_e vs. $\sqrt{43200}$ (-0.519%) and the square $(m_\mu/m_e)^2$ vs. 43200 (-1.034%) lie on *exactly the same axis* $\varepsilon_\mu - \varepsilon_e$. Logarithmically it holds exactly:

$$\ln\frac{(m_\mu/m_e)^2}{43200} = 2 \cdot \ln\frac{m_\mu/m_e}{\sqrt{43200}}.$$

The square doubles the percentage but measures no new physics. It is *one* finding, two ways of writing it. The value $-38.99 \cdot \xi$ is empirical; it lies 8σ (Eides) from $39 \cdot \xi$ and is not a Galois integer **[S]**.

Sum axis: the product is orthogonal. $\varepsilon_e \approx +18.9 \cdot \xi$ and $\varepsilon_\mu \approx -20.1 \cdot \xi$ are nearly equal in magnitude and opposite in sign. In the sum they nearly cancel; what remains is $-1.21 \cdot \xi$ — the common scale deviation of both masses. This is at order ξ, consistent with the SI projection $S^1_m \to \mathbb{R}_t$ (Docs. 085, 306, 333) **[K]**. The sum ($-1.21 \cdot \xi$) does not follow from the difference ($-38.99 \cdot \xi$) — they are two independent physical pieces of information.

Combination	Logarithm	Residual	Status
$\ln\left(m_\mu/m_e/\sqrt{43200}\right)$	$\varepsilon_\mu - \varepsilon_e$	$-38.99 \cdot \xi$	ratio residual **[K]**; cause: non-uniform ε_i (Doc. 352, § 9–10) **[K]**
$\ln\left((m_\mu/m_e)^2/43200\right)$	$2(\varepsilon_\mu - \varepsilon_e)$	$-77.99 \cdot \xi$	exactly = $2\times$ above
$\ln\left(m_e \cdot m_\mu \,/\, 54\,\text{MeV}^2\right)$	$\varepsilon_e + \varepsilon_\mu$	$-1.21 \cdot \xi$	follows from anchor + ratio residual **[K]**

The apparently different percentages -0.016%, -0.52%, and -1.03% arise from two numbers: the generation tension ($-39 \cdot \xi$) and the SI projection ($-1.2 \cdot \xi$). Their large ratio is not an amplification but a cancellation in the sum (verification script: `pruef_351_generationsdifferenz.py`).

The residual of the product comparison lies at order ξ itself:

Quantity	Residual in units of ξ	Interpretation
$m_e \cdot m_\mu$ vs. 54	$-1.21 \cdot \xi$	order ξ: SI projection **[K]**
$(m_\mu/m_e)^2$ vs. 43200	$-77.6 \cdot \xi$	ratio residual, order 100ξ **[K]**

The SI projection $S^1_m \to \mathbb{R}_t$ (Docs. 085, 306, 333) structurally produces errors of order ξ. The product residual $-0.016\,\% \approx 1.2 \cdot \xi$ lies precisely in this range and is therefore structurally explainable as an SI transition approximation **[K]**. The quadratic residual $-1.034\,\% \approx 78 \cdot \xi$ lies in higher order — ratio residual, empirically computed **[K]**.

Result: The Galois identity $m_e \cdot m_\mu = 54$ MeV^2 is the only currently known identity whose residual lies at order ξ — the structurally expected SI projection error. The residual is therefore not unexplained; it corresponds to the approximation of the SI transition.

Following Stefaan Vossen's clarification within the Constitutional Physics / Canonical Constitutional Record framework (5 September 2026): the Layer-1 object $(r_\mu/r_e)^2/\xi$ and the Layer-2 object $(m_\mu/m_e)^2_{\text{PDG}}$ are *distinguishable scientific objects*. The comparison is permitted, but the residual $\approx 78 \cdot \xi$ belongs to the Layer-1 $\to$ Layer-2 mapping, not to the algebraic identity itself.

Constitutional description (developed in correspondence with Stefaan Vossen, applying the Constitutional Physics / Canonical Constitutional Record framework, 5 September 2026)

The following four-layer formulation summarises the epistemic status of the FFGFT comparisons:

Layer	Status
Layer-1	Exact within rational arithmetic: $(m_\mu/m_e)^2 = 43200$ and $\lvert GF(3)^*\rvert \cdot \lvert GF(27)\rvert = 54$ are integer Galois identities without approximation **[B]**.
Layer-2	One empirical dimensional anchor: $m_\mu = 105.66$ MeV (PDG). All absolute predictions follow from it **[K]**.
Residual	Empirical discrepancy with a structural explanation of approximately the correct order: $(m_\mu/m_e)^2$ deviates by $\approx 78 \cdot \xi$ from the PDG value (ratio residual, order 100ξ); $m_e \cdot m_\mu$ deviates by $\approx 1.2 \cdot \xi$ (follows from anchor + ratio residual) **[K]**.
Quantitative bridge	The forward derivation of the exact residual from the $S^1_m \to \mathbb{R}_t$ non-isomorphism is not in the corpus. **Open [S].**

The open [S] is not a gap in the pejorative sense but a precisely constituted limit: the theory states explicitly where its formal construction stops.

Methodological note (developed in correspondence with Stefaan Vossen, applying the Constitutional Physics / Canonical Constitutional Record framework, 5 September 2026)

> *"Constitution does not tell the Galois identity what it is. It tells us what a particular scientific use of that identity is presently entitled to mean."*

Two distinctions to be preserved in the record:

1. **54 as a finite-field integer $\neq 54$ MeV^2 as a dimensional representation.** The identity $\lvert GF(3)^*\rvert \cdot \lvert GF(27)\rvert = 2 \cdot 27 = 54$ belongs to the dimensionless Layer-1 algebra **[B]**. $m_e \cdot m_\mu = 54$ MeV^2 is a dimensional representation produced by the mapping into SI units (Layer-2). Both are distinguishable scientific objects; the unit MeV^2 belongs to the dimensional mapping, not to the dimensionless algebra alone.
2. **Numerical correspondence vs. physical attribution.** The residual $m_e \cdot m_\mu \approx 54$ MeV^2 with deviation $\approx 1.2\cdot\xi$ is an established numerical correspondence **[K]**. The physical attribution of that residual to the SI projection $S^1_m \to \mathbb{R}_t$ (Docs. 085, 306, 333) remains bounded by the still-open quantitative bridge **[S]**. The correspondence stands; the attribution is a proposal, not a derivation.

Methodological remark: relation vs. prediction

The three percentages (-0.519%, -1.034%, -0.016%) are *not tolerance values*. Tolerance arises only when a **calculated** mass deviates from a **measured** one — which requires an independent FFGFT prediction for m_e or m_μ individually.

The Galois identities 43200 and 54 MeV^2 are **relations** between both masses, not individual predictions. As long as no independent FFGFT value for m_μ (or m_e) exists, none of the three percentages has a defined limit.

Moreover: every direct mass carries the fractal correction K_{frak} and the SI projection $S^1_m \to \mathbb{R}_t$. In the ratio, K_{frak} cancels; the SI corrections *subtract* (difference axis, $\approx -39\cdot\xi$). In the product, K^2 remains; the SI corrections *add* (sum axis, $\approx -1.2 \cdot \xi$). The different percentages arise from this — not from measurement errors.

No independent single prediction for m_μ without an input anchor exists in the corpus. With m_e as anchor (SETZUNG) there are two routes:

- **Square route:** $m_\mu = m_e \cdot \sqrt{43200} = m_e \cdot 120\sqrt{3}$ — irrational, deviation $+0.52\%$. Not a clean Layer-1 value.
- **Product route:** $m_\mu = 54\,\mathrm{MeV}^2/m_e$ — rational, deviation $+0.016\% \approx 1.2 \cdot \xi$. The only rational tolerance value for m_μ at anchor m_e.

But: $m_{e,\mathrm{PDG}}$ already carries K_{frak} and the SI projection. Hence $m_\mu^{\mathrm{pred}} = 54/m_{e,\mathrm{PDG}} \approx m_{\mu,\mathrm{bare}}(1-\varepsilon_e)$ and $m_{\mu,\mathrm{PDG}} \approx m_{\mu,\mathrm{bare}}(1+\varepsilon_\mu)$, so:

$$\frac{m_\mu^{\mathrm{pred}}}{m_{\mu,\mathrm{PDG}}} - 1 \approx -(\varepsilon_e + \varepsilon_\mu) \approx -1.2 \cdot \xi.$$

This is the sum axis — the same relation as the product residual. The fractal correction does not cancel but is contained in ε_e and ε_μ and adds in the product. A true Layer-1 tolerance value for m_μ without any correction does not exist, because every direct mass carries K_{frak} and the SI projection.

5 Error accumulation along the derivation chain

Every derivation step introduces a residual. What matters is which combination is formed, because that determines which corrections remain visible and which cancel (Section 4).

First step $m_e \to m_\mu$, rational route. With m_e as anchor (SETZUNG) and the Galois identity $m_e \cdot m_\mu = 54$ MeV^2, one obtains rationally:

$$m_\mu^{\mathrm{pred}} = \frac{54\ \mathrm{MeV}^2}{m_{e,\mathrm{PDG}}} = 105.675\ \mathrm{MeV}, \qquad \text{residual: } +0.016\,\% \approx +1.2 \cdot \xi\ \textbf{[K]}.$$

This is the sum axis ($\varepsilon_e + \varepsilon_\mu$); both masses carry K_{frak} and SI projection, which add in the sum. The irrational route $m_\mu = \sqrt{43200}\,m_e = 120\sqrt{3}\,m_e$ ($+0.52\,\%$) is not a clean Galois value (bookkeeping rule above).

Accumulation. For m_τ (next step via $m_\tau/m_\mu = 16.992$ [K], Doc. 317), for v (SM definition, not measured, R112) and α (5σ tension Rb/Cs, R113) the uncertainties of the extraction paths accumulate. To determine where a residual originates, the convergence of empirical values to algebraic ones would need to be established — it is not.

Bookkeeping: For every reported prediction state: (1) the anchor, (2) the number of derivation steps, (3) corrections applied, (4) the comparison value with extraction method. "FFGFT predicts m_μ to $0.5\,\%$" is incomplete; complete: "From m_e (PDG), in one step via $m_e \cdot m_\mu = 54$ MeV^2, m_μ follows with $+0.016\,\%$ relative to the PDG value."

σ balance of the prediction comparison. The prediction $m_\mu^{\mathrm{pred}} = 54\,\mathrm{MeV}^2/m_{e,\mathrm{PDG}}$ carries only the uncertainty of m_e:

$$\sigma(m_\mu^{\mathrm{pred}}) = \frac{54\,\mathrm{MeV}^2}{m_e^2} \cdot \sigma(m_e) = 3.1 \times 10^{-8}\ \mathrm{MeV} \quad (\sigma_{\mathrm{rel}} = 2.9 \times 10^{-10}).$$

The comparison uncertainty combines in quadrature:

$$\sigma_{\mathrm{comp}}^2 = \sigma(m_\mu^{\mathrm{pred}})^2 + \sigma(m_{\mu,\mathrm{PDG}})^2 \approx \sigma(m_{\mu,\mathrm{PDG}})^2.$$

The contribution of $\sigma(m_e)$ is $0.25\,\%$ of σ_{comp} — negligible. The entire measurement uncertainty resides in $m_{\mu,\mathrm{PDG}}$ (HFS extraction, Eides 2026). The deviation $+0.016\,\%$ lies $7386\,\sigma$ (CODATA) or $1386\,\sigma$ (Eides) outside — the same as the product residual, now expressed as a σ distance.

Classification of the product residual. The product residual $-1.2 \cdot \xi$ is at order ξ. The MeV scale is exact since 2019 (h, c, e fixed); the measurement uncertainties $\sigma_{\text{rel}}(m_e) \approx 0$, $\sigma_{\text{rel}}(m_\mu) \approx 0.0002 \cdot \xi$ do not explain the residual. It follows from the Layer-1 ↔ Layer-2 gap (near-complete cancellation of ε_e and ε_μ in the sum axis). The σ-range within which this residual is expected to lie has not been derived **[S]**.

Algebraic dependency: only one open number

With anchor m_e (SETZUNG) and the Galois ratio $(m_\mu/m_e)^2 = 43200$ (Layer-1), it follows necessarily:

$$m_\mu = m_e \cdot \sqrt{43200}, \qquad m_e \cdot m_\mu = m_e^2 \cdot \sqrt{43200} = 54.273 \text{ MeV}^2.$$

The product residual from these two inputs alone is $+37.9 \cdot \xi$ — no free parameter, no **[S]**.

The PDG product residual is $-1.2 \cdot \xi$. The difference is $39.1 \cdot \xi$ — exactly the ratio residual. This is algebraically necessary:

$$\underbrace{(\varepsilon_e + \varepsilon_\mu)}_{\text{product residual } -1.2\cdot\xi} = \underbrace{2\varepsilon_e}_{\text{from anchor } +37.9\cdot\xi} + \underbrace{(\varepsilon_\mu - \varepsilon_e)}_{\text{ratio residual } -39.1\cdot\xi}.$$

The product residual is not a third independent number — it is the sum of anchor contribution and ratio residual. There is only **one** open number: the ratio residual $-38.99 \cdot \xi$. The sole **[S]** sits there.

6 Consequences for the corpus

No numerical value in the corpus changes. What changes:

1. v **is neither anchor nor observable.** Docs. 006, 319, R104: "$v = 246$ GeV (measured)" → "$v = 246$ GeV (SM definition from G_F)". Register R112.
2. m_μ/m_e **is QED-dependent.** "model-independent" → "least model-dependent available value"; $\nu_F \propto \alpha^2$. Register R113.
3. α **has no unique measured value below 1 ppb.** Cite the comparison value as CODATA average with Rb/Cs tension.
4. m_τ**:** two methods, PDG average 1776.93(9), FFGFT 0.4σ — open.
5. **New open item [S]:** τ_μ directly from FFGFT, without detour via G_F and v — the only convention-free test of the weak sector.

Sources

All sources verified on 5 September 2026 against INSPIRE, APS, Nature, arXiv, or PubMed. Numerical values computed in exact rational arithmetic against CODATA 2022.

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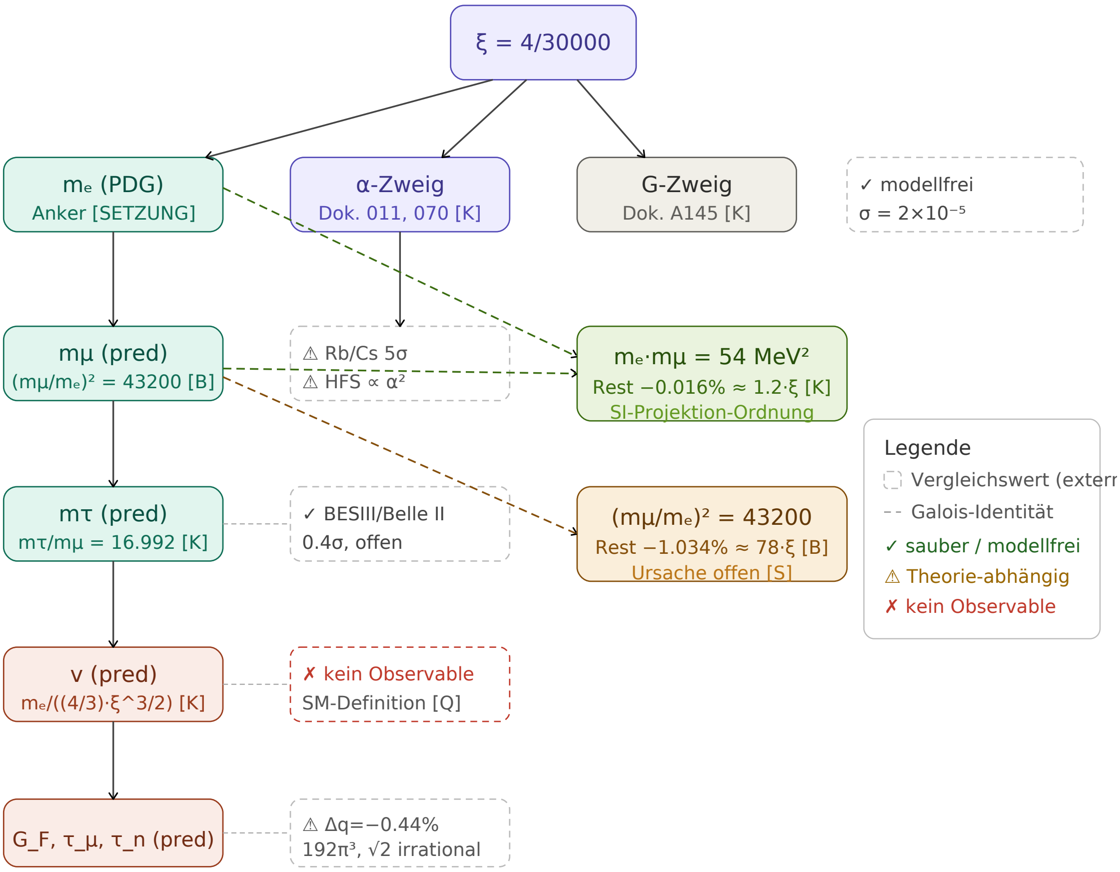

Derivation chains: from ξ to comparison values. Dashed boxes = external comparison values; dashed arrows = Galois identities.

1 Affected derivation chains

Which FFGFT chain depends on which comparison value, and what each finding changes.

Finding	Affected documents	Numerical correction	Wording correction
v is SM definition (R112)	Docs. 006, 319, R104	none	"$v = 246$ GeV (measured)" → "(SM definition from G_F)"
G_F: Δq, $192\pi^3$, $\sqrt{2}$	Docs. 006, R104, 319	none	state $\Delta q = -0.44\,\%$; clarify with/without Δq
m_μ/m_e QED-dependent (R113)	Docs. 338, 317, 011	none	"model-independent" → "least model-dependent"
α Rb/Cs 5σ	Docs. 011, 070, 338	none	cite CODATA average with tension
m_τ two methods	Docs. 317, 006	none	cite BESIII + Belle II; PDG average 1776.93(9)
Neutrinos Δm^2	Docs. 339, 340	none	freeze NuFIT configuration
G	Doc. A145	none	—
Total		**none**	wording clarifications

The residual $(m_\mu/m_e)^2 = 43200$ vs. 42753.12 (difference 446.9) is $240\,000\,\sigma$ (CODATA) or $45\,000\,\sigma$ (real uncertainty, Eides) of the measurement uncertainty — empirical discrepancy **[K]** (Doc. 352, § 9–10); physical attribution (R113 bridge) open **[S]**. No numerical value in the corpus changes.